



Contents lists available at ScienceDirect

Clinical Nutrition ESPEN

journal homepage: <http://www.clinicalnutritionespen.com>

Original article

Real-world experience with glucagon-like peptide 2 analogues in patients with short bowel syndrome and chronic intestinal failure: Results from an international survey in expert intestinal failure centers



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ARTICLE INFO

Article history:

Received 2 April 2024

Accepted 27 October 2024

Keywords:

Short bowel syndrome

Intestinal failure

Glucagon-like peptide 2 analogue

Teduglutide

Glepaglutide

Real-world data

SUMMARY

Background and aims: Glucagon-like peptide 2 (GLP-2) analogues are the first available disease-modifying treatments for patients with intestinal failure (IF) due to short bowel syndrome (SBS). Efficacy in terms of reduction of parenteral support (PS) has been demonstrated in multiple studies and real-world reports. However, it remains unclear how many patients are eligible to receive the treatment, when treatment is started after intestinal resection, how treatment efficacy is assessed outside of clinical trials, and how the treatment is modified in case of non-response or adverse events. The aim of this study was to investigate the real-world management of patients treated with GLP-2 analogues in expert centers around the world.

Methods: A survey questionnaire was developed by a multidisciplinary working group consisting of 52 questions related to various aspects of multidisciplinary care of SBS-IF patients. The 17 questions related to the use of GLP-2 analogues in clinical practice were analyzed for this study. The online survey was sent to 33 participating centers in a phase 3 study of a long-acting GLP-2 analogue. Only responses from countries with access to commercially available GLP-2 analogues were included in the study. A descriptive analysis was performed for each question. Results are presented as median (interquartile range).

Results: The responses from the 19 expert IF centers with access to GLP-2 analogues indicated that 10 (10–20) % of patients with SBS-IF were treated with a GLP-2 analogue, which was less than the number of eligible patients (30 (25–40) %). In most centers (10 centers, 53 %), GLP-2 therapy was started 6–12 months after the last intestinal resection, with 5 centers (26 %) starting later (12–24 months). Multiple parameters were used in combination to determine the response to GLP-2 analogues of which the three

Abbreviations: EASE, Efficacy And Safety Evaluation; GLP-2, Glucagon-like Peptide-2; IF, Intestinal Failure; MDT, Multidisciplinary Team; MWG, Multidisciplinary Working Group; PPI, Proton-Pump Inhibitor; PRO, Patient-Reported Outcome; PS, Parenteral Support; PYY, Peptide YY; SBS, Short Bowel Syndrome; US, United States.

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<https://doi.org/10.1016/j.clnesp.2024.10.161>

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most common were >20 % decrease in PS (95 %), at least 1 day of PS reduction per week (84 %) and increased urinary output (68 %). In non-responders GLP-2 therapy was stopped within the first year by 67 % of the centers. Finally, strategies in case of significant adverse events include stopping the GLP-2 analogue (used by 79 % of experts), dose reduction (67 %) and temporary treatment interruption (62 %). *Conclusion:* The results of this survey completed by expert IF centers show the real-life use of GLP-2 analogues in clinical practice. Key learning points identified include the accounting for a period of intestinal adaptation before starting GLP-2 analogues and not stopping the treatment too early in case of non-response. The best strategy in case of adverse effects should be studied further.

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1. Introduction

Short bowel syndrome (SBS) is a rare condition in which intestinal resections have resulted in a small bowel length in continuity of less than 200 cm in adults [1]. Intravenous supplementation of fluids, nutrients, and/or electrolytes is necessary if the reduced surface area falls below the minimal threshold to maintain adequate absorption to sustain a healthy homeostasis and growth in children. This condition is defined as intestinal failure (IF) [2]. IF is the rarest form of organ failure with a prevalence of 5–80 per million [2,3]. The treatment of patients with SBS-IF is multimodal and requires the input of an interdisciplinary team with clinical expertise in the field. Therefore, it is recommended that the care for these complex and heterogeneous patients is centralized in expert centers with a multidisciplinary team (MDT) consisting of physicians, surgeons, nurses, dietitians, pharmacists, and others with expertise in IF [1,4,5].

By definition patients with SBS-IF are dependent on parenteral support (PS) which is composed of fluids with or without nutrients and electrolytes. The PS regimen should be individualized to the specific needs of the patient and take into account the remnant gastrointestinal anatomy, oral intake, and intestinal losses [2,6]. The ultimate patient goal is to achieve enteral autonomy, i.e. weaning off PS, which is possible in about half of the patients with SBS-IF within 5 years through a process of intestinal adaptation [7]. This adaptive response can be supported by nutritional and medical therapy. In the remaining 50 % of patients who remain dependent on PS, the goal is to reduce PS requirements, avoid IF- and therapy-related adverse events, and improve quality of life. Pharmacologic therapy consists of anti-secretory (e.g., proton-pump inhibitors (PPI)) and anti-motility agents (e.g., loperamide) to reduce gastrointestinal secretions and slow transit [1,8]. Dietary modifications can aid in reducing intestinal losses, stimulating absorption of fluids and nutrients, and promoting the natural intestinal adaptation process, especially in patients with a colon-in-continuity [9]. The adaptation process involves the secretion of pro-adaptive hormones, including glucagon-like peptide 2 (GLP-2), GLP-1, peptide YY (PYY) and likely others [10–13].

In the last decade, analogues of the naturally occurring GLP-2 have revolutionized the treatment of SBS-IF [10]. Clinical response rates – at least 20 % reduction of PS at week 20 and 24 – of 63 and 66 % were reported for the short-acting analogue teduglutide and the long-acting glepaglutide [14,15]. At the time of the survey only teduglutide was commercially available. Several long-acting GLP-2 analogues are currently in different phases of clinical development for SBS indication. However, due to the heterogeneity of the SBS population, optimal use of GLP-2 analogue therapy in clinical practice remains challenging. The aim of this

study was to investigate the real-world management of patients treated with GLP-2 analogues in expert centers around the world.

2. Materials and methods

2.1. Survey

An international multidisciplinary SBS IF management working group (MWG) was established in 2020. The MWG comprised of SBS experts across various disciplines, including physicians (gastroenterologists and surgeons with expertise in IF), dietitians, a nurse, a pharmacist and a nutritionist. The primary goal of the MWG was to identify remaining gaps in the management of SBS through a literature search followed by an online survey to further explore the identified gaps in real-life clinical practice. The development process has recently been described in detail [16,17]. The survey questionnaire consisted of 52 questions related to the composition of the SBS-IF multidisciplinary team [4], methods used for managing high-stool output associated with SBS [18] and use of GLP-2 analogues in clinical practice. The portion of the questionnaire addressing the use of GLP-2 analogues consisted of 17 questions and is the focus of this manuscript.

2.2. Participating centers and data collection

All IF centers participating in the Efficacy And Safety Evaluation (EASE) phase 3 clinical trials of glepaglutide, a long-acting GLP-2 analogue for treatment of SBS [19], were invited to participate in this survey. A total of 33 referral IF centers (United States (US) 11, Germany 5, United Kingdom 5, Canada 3, Poland 3, Denmark 2, France 2, Belgium 1, Netherlands 1) were surveyed.

The online questionnaire was created in SurveyMonkey, and a link was distributed directly to the principal investigators for the EASE trial in January 2021. Each center was asked to answer the survey questions as one MDT. The questions regarding patient characteristics were specified to adult patients (≥ 18 -year-old) with a diagnosis of chronic IF. The center's responses to the questionnaire were collected anonymously.

This study was approved by the research ethics committee of the University of Alberta, Canada (Pro00107599; Principal Investigator Dr. Leah Gramlich).

2.3. Statistical analysis

Only responses from countries where IF teams had access to teduglutide – the only marketed GLP-2 analogue at the time of the survey – were evaluated. A descriptive analysis was performed for each question. Results are given as median (interquartile range)

unless specified otherwise. Where applicable, a separate analysis was performed for US vs. non-US centers.

3. Results

3.1. Participating centers

Thirty out of the 33 centers (91 % response rate) completed the survey. At the time of the survey only the US (9 centers, 47 %), Canada (3 centers, 16 %), Germany (5 centers, 26 %) and France (2 centers, 11 %) had access to teduglutide (Gattex® in North-America or Revestive® in Europe and Canada; Takeda, Japan). Therefore, only the responses of these 19 centers were included in the analysis.

4. Patient selection and funding

The proportion of patients with SBS-IF fulfilling the criteria to start a GLP-2 analogue, as assessed by the MDT at each of the 19 centers, taking into account the label, local regulations and personal experience - was estimated at 30 (25–40) %, but ranging from 10 to 90 %. However, of those patients with SBS considered appropriate candidates for GLP-2 analogue therapy, only 10 [10–20] % were actually treated with GLP-2 analogues.

In 5 out of the 10 non-US centers, funding was almost exclusively provided by publicly funded healthcare. In contrast, private and public insurance companies were funding GLP-2 analogues in the US centers. When asked about economic barriers to prescribing GLP-2 analogues, 5 US centers (56 %) and 4 non-US centers (40 %) mentioned the lack of coverage by the funding agency in a variable proportion of patients (20–100 %), while high patient co-payment was only an issue in the US (6 centers, 67 %).

4.1. Initiation of GLP-2 therapy and follow-up

In most centers (10 centers, 53 %), GLP-2 therapy was started 6–12 months after the last intestinal resection leading to SBS-IF; with 5 centers (26 %) starting later (12–24 months) (Fig. 1). No center reported starting GLP-2 analogue therapy earlier than 6 months after intestinal resection leading to SBS-IF.

Patient/caregiver training of GLP-2 analogues was performed by a variety of individuals, but the most common response was a nurse paid by the pharmaceutical company (11 centers, 58 %; 6 US centers

and 5 non-US centers). Other specialties performing GLP-2 training included a home PN nurse (5 centers, 26 %), intestinal failure physician (2 centers, 11 %), dietitian, home care company or a pharmacist (1 center each).

After initiation of a GLP-2 analogue, the first follow-up was scheduled after 1–2 weeks in most centers (11 centers, 58 %), either virtually or in person (Fig. 2). One third of the centers performed the first check within the first week following initiation of GLP-2 therapy. Patients who are considered stable on GLP-2 therapy are evaluated at 3–6 month intervals in most centers (11 centers, 58 %), while 7 centers (37 %) continue follow-up visits with intervals of less than 3 months.

4.2. Response to GLP-2 therapy

A variety of parameters were used in combination to determine the response to GLP-2 analogues (Fig. 3). The three most cited positive outcome measures were >20 % decrease in PS (95 %), at least 1 day of PS reduction per week (84 %) and increased urinary output (68 %). In those patients considered by the MDT to be non-responders, GLP-2 therapy was stopped within the first year at most centers (12 centers, 67 %) (Fig. 4).

4.3. Adverse events

The four most common early (<6 months after initiation of therapy) adverse events were abdominal pain (17 centers, 89 %), hypertrophy of the stoma (16 centers, 84 %), injection site reactions (11 centers, 58 %) and fluid overload (11 centers, 58 %), encountered in a variable proportion of patients (Suppl Fig. 1A). The three long-term (beyond 6 months of therapy) adverse events that were most often mentioned by the centers were hypertrophy of the stoma (12 centers, 63 %), abdominal pain (11 centers, 58 %) and biliary adverse events (7 centers, 37 %) (Suppl Fig. 1B). In case of significant adverse events, the most common strategy was to stop the GLP-2 analogue (15 centers, 79 %), but also dose reduction (14 centers, 67 %) and resuming therapy at a lower dose after a short interruption of the treatment (13 centers, 62 %) were commonly used. Prolonging the dosing interval was used less frequently (4 centers, 21 %).

The most common reasons to stop GLP-2 analogues by expert IF centers were patient-reported adverse events (19 centers, 100 %), patient's decision (16 centers, 84 %), lack of response (15 centers, 79 %), and lack of compliance (12 centers, 63 %).

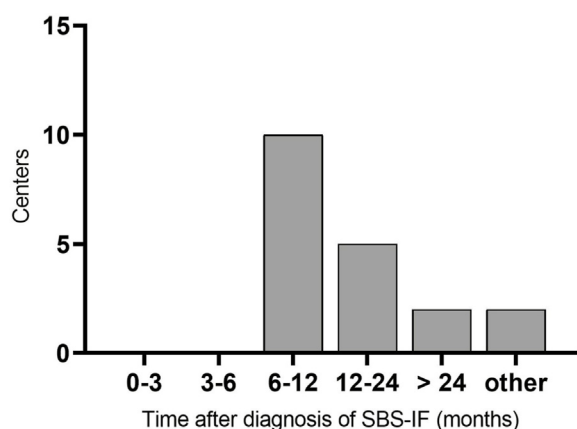


Fig. 1. Timing of initiation of GLP-2 analogues after bowel resection leading to the diagnosis of SBS-IF. GLP-2: glucagon-like peptide 2; SBS-IF: short bowel syndrome with intestinal failure. Other: variable duration (1 center) and patient parameters other than time (1 center).

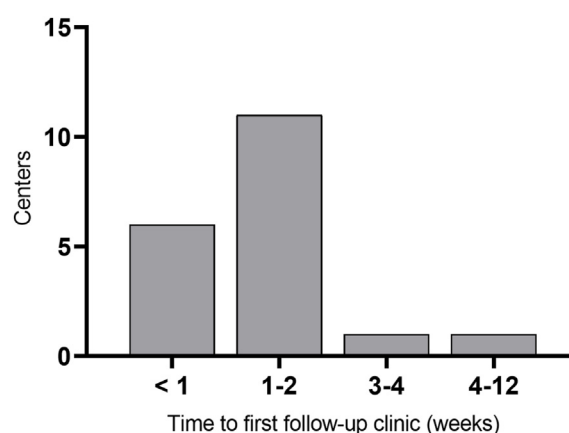


Fig. 2. Timing of the first follow-up visit (virtual or in person) after initiation of GLP-2 analogues. GLP-2: glucagon-like peptide-2.

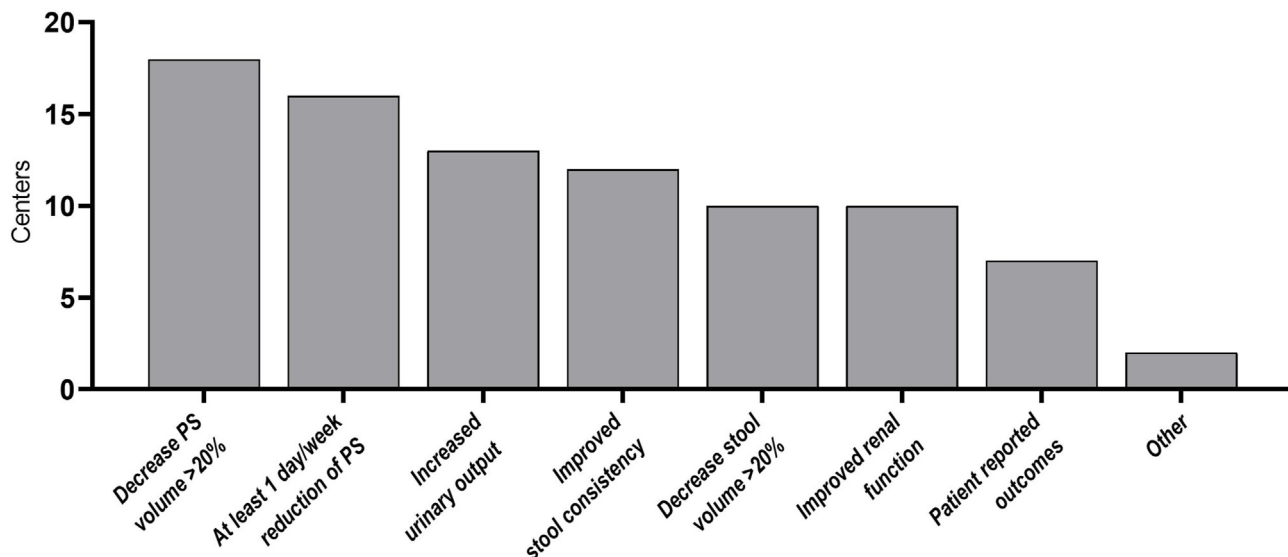


Fig. 3. Parameters used to determine clinical response to GLP-2 analogues (multiple answers per center possible). Other: weight gain (1 center) and reduced thirst (1 center). GLP-2: glucagon-like peptide 2, PS: parenteral support.

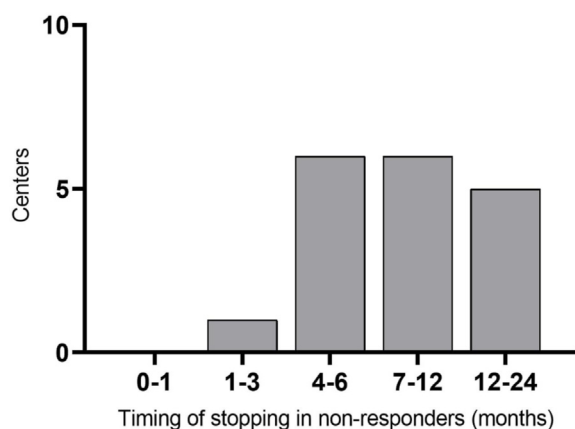


Fig. 4. Timing of stopping GLP-2 analogues in patients who are considered non-responders. GLP-2: glucagon-like peptide 2.

5. Discussion

GLP-2 analogues are the first available disease-modifying treatments for patients with SBS-IF [20]. Multiple studies (including phase 2, 3 and open-label extension studies) as well as real-world data have demonstrated the efficacy of the short-acting GLP-2 analogue teduglutide in reducing the need of PS [10]. However, it was still unclear how many eligible patients are actually receiving the drug, when treatment is started after intestinal resection, how treatment efficacy is assessed outside of clinical trials, and how the treatment is modified in case of non-response or adverse events. In the current survey, the respondents from expert IF centers indicated that only about 10 % of patients with SBS-IF were treated with a GLP-2 analogue, which was less than the number of eligible patients. When a GLP-2 analogue was started, the majority of patients were initiated relatively early, i.e. within the first year after the last major intestinal resection. In contrast to clinical trials, response was not assessed by one criterion in most centers, but by a combination of parameters, including >20 % reduction of PS, having at least one additional day off PS and increased urinary output. In case of non-response to GLP-2

analogues, therapy was stopped as early as within the first year after initiation by 2/3 of the centers. Finally, even though most experts would stop GLP-2 analogues in case of significant adverse events, other strategies such as dose reduction and temporary treatment interruption were used as well.

Teduglutide was approved by the FDA and EMA in 2012 [20]. Nevertheless, at the time of the survey in 2021, only 19 out of the 30 expert centers had access to teduglutide in clinical practice. At the time of the survey glepaglutide and apraglutide were not commercially available but phase 3 trials were ongoing. Even if the other 11 centers had experience with glepaglutide, apraglutide and/or teduglutide in clinical trials, they were not included in the current analysis because the responses were thought to reflect the study protocol and not its use in everyday clinical practice. This observation highlights the inequality in SBS-IF care between regions [21]. One barrier to prescribing GLP-2 analogues is the high cost, identified by 56 % of US and 40 % non-US centers. With more competitors entering the market, including long-acting GLP-2 analogues, lower drug costs are anticipated, which should allow increased access and use of GLP-analogues for therapy of SBS-IF in more countries [10].

A second important observation was the low percentage (10 %) of patients with SBS-IF treated with a GLP-2 analogue, despite 30 % of patients thought by the expert center to fulfill the patient selection criteria for use. In this survey, the eligibility criteria were not defined and left to the discretion of the respondent. One possibility is that the experts interpreted the criteria as the defined medical conditions recommended to start a GLP-2 analogue, i.e. SBS-IF after a sufficiently long period of adaptation on a stable regimen of PS and without contraindications such as a recent malignancy or congestive heart failure with fluid retention [1,6,8]. The survey response of 30 % of patients meeting GLP-2 analogue eligibility corresponds quite nicely with a report from a single expert center in Italy where 34 % of patients with SBS-IF had absolute and 30 % relative contraindications, leaving 36 % of patients as good candidates for GLP-2 analogues from a medical point of view [22]. However, other factors should be considered, including predicted adherence, acceptance of the treatment and/or concern for adverse events by the patient, and presence of social support. Indeed, a recent study from Paris indicates that non-medical issues accounted for 18 % of the reasons why certain patients were not treated

with a GLP-2 analogue. This likely explains the discrepancy between patient eligibility and those actually treated with a GLP-2 analogue in the current survey [23]. Moreover, the latter study also shows difference between the prevalent SBS patient population when teduglutide became available in France in 2015 vs. the so-called incident population which were new patients diagnosed with SBS-IF after this point in time. In the prevalent population, 28 % of patients with SBS-IF were treated with teduglutide vs. only 8 % in the incident population. The incident population was also characterized by a longer remnant bowel length, suggesting a greater potential of natural adaptation which also translated into higher success rates of teduglutide in terms of PS volume reduction [23].

The ESPEN guidelines for chronic IF recommend starting GLP-2 analogues after a period of adaptation of 12–24 months [6], similar to the phase 3 teduglutide study where an adaptation period of 12 month after last intestinal resection was required [14]. However, even if none of the experts in the current survey indicated that they would start before 6 months, 53 % of centers responded that they would start between 6 and 12 months and 26 % between 12 and 24 months. While the questionnaire lacked granularity to know exactly how early this was (i.e., month 6 vs. month 11), it is clear that IF experts tend to start GLP-2 analogues earlier than what is recommended in the guidelines and indicated on the label in some countries [21]. One of the reasons for earlier treatment initiation may be the concept of accelerated (hyper)adaptation, which refers to an adaptation process boosted by e.g. GLP-2 analogues [8,24]. The effect of early introduction of GLP-2 analogues is still unclear, even if real-world data indicate that earlier treatment – but still longer than 12 months after diagnosis of SBS – has a superior efficacy than later treatment in long-standing SBS patients [23]. In a recent phase 3 study with the long-acting GLP-2 analogue glepaglutide patients were eligible for inclusion from 6 months after the last resection with similar outcome compared to the teduglutide studies, which would support earlier initiation of the treatment [15]. The access and costly nature of the treatment are other barriers to treat patients early, who could potentially wean off PS spontaneously over time, especially patients with a colon-in-continuity [25]. Finally, it is to be expected that the first patients treated after market access will be the more stable patients with long-standing SBS-IF, while the average duration between resection and starting a GLP-2 analogue will become shorter once the treatment is available for a longer period.

Most of the centers check on their patients, either virtually or in person, within 2 weeks of starting a GLP-2 analogue and 1/3 of centers even after the first week. This implies that expert centers anticipate an early response to therapy. This may be especially true for patients with an end-jejunostomy [25], which may lead to adverse events like acute fluid retention necessitating aggressive PS volume reduction. It is also reasonable to have early and frequent follow-up for the first three months following initiation of therapy in patients without a colon-in-continuity [25]. Once patients reach a new stability under GLP-2 analogue treatment, regular 3-month follow-ups were frequently utilized in most expert centers.

The classical primary endpoint (defined as “clinical response”) to GLP-2 analogue therapy used in clinical trials was at least a 20 % reduction in PS volume [14]. The same criterion for clinical response was also used in the reporting of real-world data [26–28]. The origin of this arbitrary clinical response is based on an average PS requirement of 5 days per week whereas a 20 % volume reduction would equate to 1 less day of PS per week, which is thought to be clinically significant. Like the clinical trials, this was the most used definition of clinical response identified in this real-life survey.

Other parameters considered when assessing clinical response in real-life practice by this survey includes reduction of PS frequency, increased urinary output, improved stool consistency, and patient reported outcomes (PRO). Recently, a 20 % PS volume reduction was validated as important in terms of improvement of PRO in a phase 3 study on efficacy and safety of the long-acting GLP-2 analogue glepaglutide [15]. The full data including efficacy and safety endpoints in the registrational phase 3 trials of the long-acting analogues glepaglutide [15] and apraglutide [29] are not yet available. Therefore, it remains to be seen how these different GLP-2 analogues, following regulatory approval, will be implemented in clinical practice. Quality of life (QoL) was not commonly used as a response criterion by the participants. Moreover, quality of life – assessed by the SBS-QoL questionnaire – was not significantly better in teduglutide-treated patients vs. placebo in the phase 3 study [30]. Nevertheless, using validated SBS-specific QoL-questionnaires may help the clinician to manage the patient and generate insightful real-world evidence on QoL following GLP-2 analogues therapy.

For patients who do not respond to short-acting GLP-2 analogues, 2/3 of the centers would stop the therapy within the first year and 7 out of 19 centers (37 %) within the first six months. Based on a post-hoc analysis of the STEPS phase 3 study and the STEPS-2 open label extension study, only 7 out of 34 responders (20 %) achieved at least a 20 % PS volume reduction beyond 6 months of treatment, all of whom had a colon-in continuity [25]. Therefore, clinical response is less likely to occur beyond 6 months in patients with an end-jejunostomy but may be delayed in patients with a colon-in-continuity.

Finally, adverse effects and their impact on GLP-2 analogue treatment were assessed. Changes in stoma appearance and abdominal pain were the two most frequent early and late adverse events, similar to what was observed in clinical trials and real-world cohort studies [27,31]. Injection site reactions were only notable as early adverse events by 58 % of the respondents. There is currently no clear guidance how to manage significant adverse events experienced with GLP-2 analogues, as shown by the variety of practices utilized in this survey. While stopping or holding treatment was the most common practice, centers also reduced treatment intensity by decreasing the dose per day or prolonging the dosage interval (e.g. from daily to every other day). This survey did not correlate the approach taken with the nature of the adverse event. For example, prolonging the injection interval from daily to every other day seems preferable for injection site reactions, but it is unclear whether this is superior to daily injections of a reduced dose in terms of maintaining a treatment response. Risk of developing intestinal ‘cancer and pancreatitis’ following long-term treatment with GLP-2 analogues were mentioned by 4 and 5 sites, respectively. It is unclear whether these responses were based on real-world experience at the sites and/or theoretical considerations based on the hyperplasia mode-of-action of GLP-2 analogues and the regulatory label of teduglutide (Gattex-Revestive®). Indeed, endoscopic (and/or radiological imaging) procedures for screening and monitoring GLP-2 treated patients remain important parts of standard-of-care. While this is the first survey assessing real-life practice regarding treatment with GLP-2 analogues among IF expert centers around the world, certain limitations should be acknowledged. First, only 19 centers were included in the study, based on participation in a phase 3 study of a long-acting GLP-2 analogue and access to the short-acting teduglutide at the time of the questionnaire. Second, based on the selection criteria, only larger and more experienced centers were included, potentially leading to a selection bias. Third, the survey questions were not

designed to assess for differences in practice according to the post-surgical GI anatomy (i.e. colon-in-continuity vs. end-jejunostomy). Finally, even though the questionnaire aimed to assess real-life practice, there is potential that centers responded to questions with the desirable or 'correct' response rather than their real-life practice.

6. Conclusion

Analysis of this survey completed by expert IF centers depicts the real-life use of GLP-2 analogues in clinical practice. Key learning points identified include the practice of accounting for a period of intestinal adaptation before starting GLP-2 analogue therapy and not stopping the treatment too early in case of non-response, especially in patients with a colon-in-continuity. It also highlights opportunities for future research, including how best to adjust GLP-2 analogue treatment in the setting of adverse effects.

Author contribution

Conceptualization: All authors - Formal Analysis: TV - Methodology: All authors - Project Administration: MB-H - Resources: MB-H - Supervision: LG, MB-H, - Writing – original draft: TV - Writing – review and editing: all authors.

Funding statement

Zealand Pharma provided administrative support for the management of the working group and support to set up the online survey. None of the authors received funding for the current study.

Declaration of competing interest

Tim Vanuytsel has served as a consultant and/or speaker for Abbott, Baxter Healthcare Corporation, Fresenius Kabi, Takeda Pharmaceuticals, VectivBio, and Zealand Pharma and has received research funding from Takeda Pharmaceuticals and VectivBio. Sophie Greif has served as a consultant and/or speaker for Takeda Pharmaceuticals. Elizabeth Wall has served as a consultant and/or speaker for Zealand Pharma. Hilary Catron has served as a consultant for Baxter Healthcare Corporation. Vanessa Kumpf has served as a consultant for Fresenius Kabi, Takeda Pharmaceuticals, and VectivBio. David Mercer has served as a consultant for Takeda Pharmaceuticals and Zealand Pharma. Mark Berner-Hansen is employed by Zealand Pharma. Leah Gramlich has served as a consultant and/or speaker for Abbott, Fresenius Kabi, Takeda Pharmaceuticals, and Baxter Healthcare Corporation.

Acknowledgments

Zealand Pharma provided administrative support for the management of the working group and support to set up the online survey. We thank biometrician Kim Mark Knudsen, Zealand Pharma, for assistance with data management, Christina Belza (Transplant and Regenerative Medicine Center, Toronto (ON), Canada) for providing comments to the manuscript and Adela Delgado (Royal Alexandra Hospital, Alberta Health Services, Edmonton (AB), Canada) for their help with the design of the survey.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.clnesp.2024.10.161>.

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